James Stainer

Machine learning for medicine. London, UK.

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Experience

`Dec 2023 to present`

Freelance AI Evaluator and Data Science Consultant

Remote

AI evaluation and data science across multiple client projects: model-output review, scientific reasoning benchmarks, failure-mode analysis and applied AI workflows. I write hard bioinformatics and scientific coding tasks that benchmark how frontier LLM agents reason, then audit the agent transcripts for answer leakage, shortcuts and data bugs.

`Oct 2022 to Apr 2023`

Data Science Intern

AstraZeneca, Centre for Genomics Research, Cambridge

Built a PubMed-to-Elasticsearch pipeline indexing over 35M biomedical citations and abstracts. Added NLP pipelines, using BioWordVec embeddings, that ranked candidate drug-target-disease relationships for expert review in place of broad manual search.

`Jun 2020 to Sep 2020`

Data Science Intern

AstraZeneca, Centre for Genomics Research, Cambridge

Co-developed the DrugnomeAI models, working across the ML pipeline from data collection and feature engineering through to training, validation and benchmarking.

Selected work

`PhD`

Forecasting vital signs in paediatric intensive care

UCL and Great Ormond Street Hospital

Used a PyTorch StemGNN model to forecast heart rate and blood pressure together from 5-second bedside-monitor data. Built the clinical time-series pipeline, from 4,120 raw admissions to 364 high-quality patients and about 12.5M observations. Reached 2.30% and 3.49% MAPE on 5-minute-ahead forecasts at 1-minute resolution, against 2.82% and 3.85% for ARIMA.

`2022`

DrugnomeAI

Published framework for predicting gene druggability

Scores all 19,846 human protein-coding genes using 324 features from 15 sources, with specialised models by drug modality and therapeutic area. Outperformed three published benchmarks.

Education

`2021 to 2026`

PhD, AI-enabled Healthcare

University College London

Viva passed October 2026 with minor corrections.

`2020 to 2021`

MRes, AI-enabled Healthcare

University College London

`2019 to 2020`

MSc, Data Science

Lancaster University

`2016 to 2019`

BSc (Hons), Mathematics

Lancaster University

Publications and presentations

2024	Poster, IEEE International Conference on Biomedical and Health Informatics (BHI)
2024	Poster, IEEE Engineering in Medicine and Biology Society Conference (EMBS)
2022	**DrugnomeAI is an ensemble machine-learning framework for predicting druggability of candidate drug targets** Raies, Tulodziecka, Stainer, et al. *Communications Biology* 5, 1291

Skills

Languages	Python, SQL, R, Bash
Machine learning	PyTorch, TensorFlow, scikit-learn, graph neural networks, NLP, ensemble methods
Methods	Time-series forecasting, statistical and Bayesian inference, change-point detection
Data	Data pipelines, pandas, NumPy, Elasticsearch, feature engineering, SLURM, Git